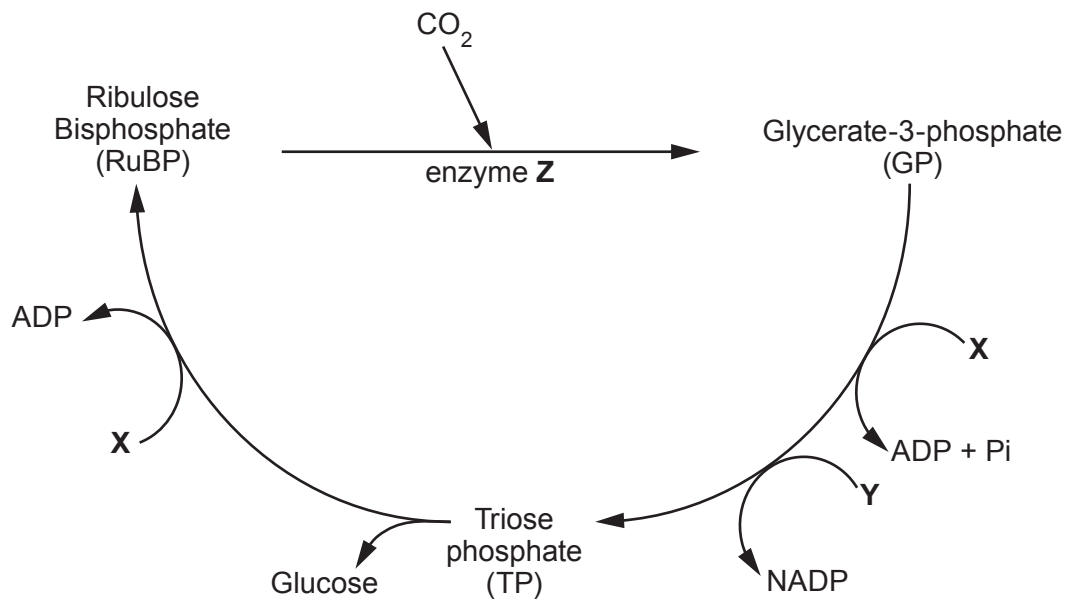


Answer all questions.

1. Tomatoes are an important food crop that can be grown in commercial greenhouses. The greenhouses often have cooling fans.

When a tomato plant is exposed to light the following reactions take place in the stroma of a chloroplast.



- (a) (i) Identify substances **X** and **Y**. [1]
- X**
- Y**
- (ii) Name enzyme **Z**. [1]
-



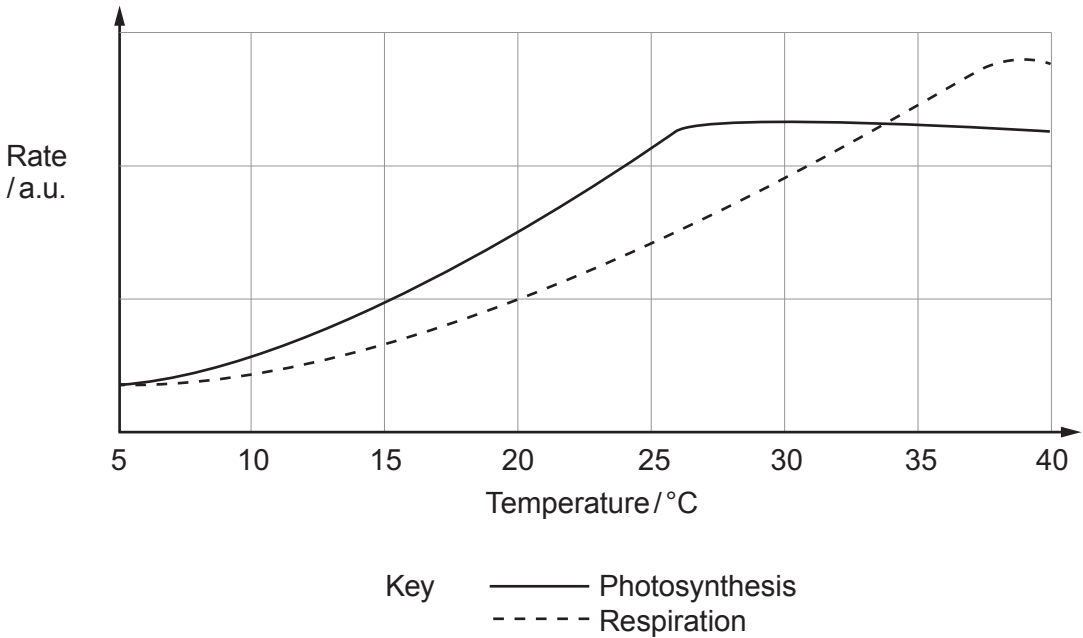
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The graph below shows the relative rates of photosynthesis and respiration at increasing temperature.



(c) With reference to the graph, explain why tomato plants grown in a greenhouse at 27°C produce sweeter tasting fruit than those grown at 40°C. [3]

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